OSA Data Assessment Report

Personal details	**Gender:**	F	**Age:**	50	**Weight & Height:**	82 kg (180.8 lbs) 163 cm (5'4.2")
					BMI:	30.9

Clinical background: Hyperlipidemia, well-controlled hypertension, menopause with hormone replacement therapy. History of radical conchotomy (2002). Multiple allergies managed with antihistamines and nasal steroids.

Medications: Sertraline, Zilergy, occasional intranasal steroid sprays.

ENT and FOL Findings: Mallampati class 2, +1 tonsils, post-orthodontic retrognathia and micrognathia, with circular velopharyngeal narrowing. FOL revealed medial wall collapse near the molars and a hypertrophic tongue base (grade 2–3), likely of muscular origin. The epiglottis and vocal cords appear normal. Significant airway improvement was noted with mandibular advancement. No septal deviation; post-radical turbinate reduction evident.

Patient presentation: The patient reports loud snoring, unrefreshing sleep, and waking up feeling tired. She experiences significant daytime fatigue, particularly around noon, and reports episodes of gasping for air and witnessed apnea. She occasionally dozes off unintentionally during passive situations and has difficulty maintaining alertness during tasks requiring attention, including driving. No TMJ issues, bruxism, or sedative use reported. Alcohol consumption is occasional.

Patient goals: Snoring cessation, reduction in nighttime choking sensations, improved daytime alertness and energy levels, better sleep quality, and reduction of health risks associated with sleep apnea.

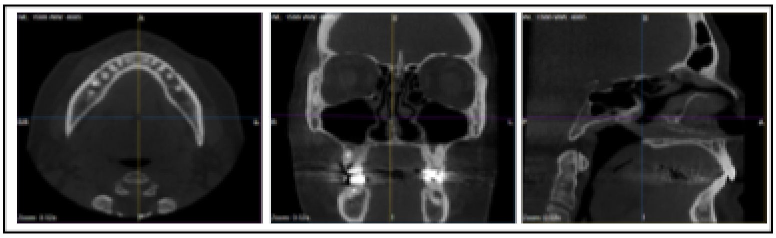
Sleep Study Data

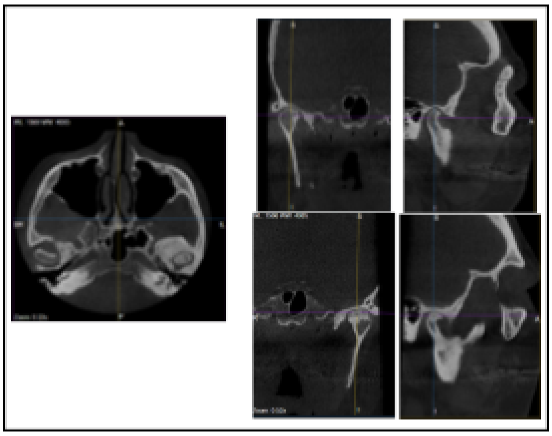
AHI	15 (Mild to Moderate OSA)	**RDI**	16.1
Supine AHI	16.5 (55% of sleep time)	**Supine RDI**	17.1
Rt. AHI	30.7 (8% of sleep time)	**Rt. RDI**	34.8
O_2 Nadir	83%	**ODI**	6.7
Less than 90% O_2	0.1%	**Supine ODI**	8.7
Snoring	mean 53 dB, >40 – 68%, >60 – 37%	**Rt. ODI**	10.2
Sleep efficiency	91%		

Observations:

- Data indicate mild to moderate OSA (AHI 15), with a slightly higher AHI of 16.5 in the supine position (55% of sleep time), and significantly elevated AHI of 30.7 when sleeping on the right side (8% of sleep time).
- Desaturation events were recorded with an ODI of 6.7; oxygen saturation dropped to a minimum of 83%, though time spent below 90% was minimal (0.1%).
- Moderate to loud snoring was observed, with mean intensity of 53 dB — present during 68% of sleep time above 40 dB and 37% above 60 dB.

Structural Observations from Imaging Data





Key Observations	Details
Obstruction Sites	Velopharyngeal, oropharyngeal, and tongue base restriction.
Soft Palate & Uvula	Elongated and swollen, partially obstructing the oropharynx.
Tongue Position	Large, posteriorly positioned tongue base encroaching the airway.
Bite & Jaw Structure	Excessive overjet and overbite, and retruded mandible.
Hyoid Bone	Not seen
Nose & Sinuses	Post-radical turbinate reduction noted.
TMJ	Degenerative changes in both temporomandibular joints. • **Right TMJ**: The condyle appears posteriorly positioned with flattening of the articular surface and cortical irregularities, consistent with joint remodelling. • **Left TMJ**: The condyle exhibits abnormal morphology with pronounced contour deformity and cortical discontinuity. Subcortical radiolucency may reflect remodelling processes or pressure-induced bone resorption, raising concern for more advanced structural alteration.

Key Observations	Details
	Joint Space: Narrowing of the intra-articular space is observed bilaterally. **Temporal Bone (Glenoid Fossa):** The temporal component appears thickened beyond normal limits, with a ground-glass appearance and altered bony architecture.

Conclusion: The provided CBCT imaging suggests potential airway obstruction primarily due to anatomical factors — including soft tissue hypertrophy and craniofacial structure. Degenerative changes are observed in both temporomandibular joints, with abnormal morphology of the left condyle and bilateral joint space narrowing, which may influence oral appliance therapy planning. Additionally, thickening and altered architecture of the temporal bone adjacent to the glenoid fossa may indicate an underlying pathology such as fibrous dysplasia. Further imaging — including MRI — is recommended to assess the extent of joint involvement and rule out significant structural abnormalities and underlying pathology.

Possible Treatment Considerations

- Given the degenerative changes in both temporomandibular joints — particularly the abnormal morphology of the left condyle and the cortical irregularities observed — as well as the thickening and altered architecture of the adjacent temporal bone, oral appliance therapy may impose additional stress on the TMJs and surrounding structures, and is not recommended at this stage.
- CPAP therapy is currently the preferred treatment option and should be initiated following mask fitting and in-lab titration to determine optimal settings.
- If CPAP intolerance occurs and a maxillofacial specialist confirms that oral appliance therapy is safe, mandibular advancement may help improve upper airway patency and stability. Additionally, anterior and posterior vertical opening can assist in creating more space for posterior tongue repositioning and increasing the distance between the soft palate and the pharyngeal wall — potentially enhancing airway patency.

Device Design Data Considerations

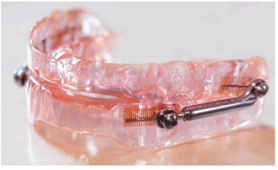
Parameter	Data-Based Consideration
Mandibular Advancement	Edge-to-edge positioning
Vertical Opening	4mm anterior
Anterior Window	Large, allowing tongue access to upper incisors

Retention Features	Ball clasps for future adjustments, elastic hooks for stabilization
Material	High-density acrylic (hygienic, adjustable)
Anterior Acrylic	Full coverage
Coverage	Full coverage
Clinical Notes	During the initial weeks of oral appliance use, it is recommended to avoid elastic bands in order to preserve natural mandibular downward movement and allow for maximal tongue advancement. After approximately four weeks, if snoring persists, elastic bands may be introduced selectively to improve lip seal and reduce residual snoring, based on the patient's response and tolerance.

Recommendations for Further Evaluation

- A comprehensive evaluation by an oral and maxillofacial specialist is recommended to assess the extent of temporomandibular joint degeneration prior to initiating oral appliance therapy.
- Consider MRI imaging of the temporomandibular joints (open and closed mouth) to evaluate disc position, joint function, and to rule out significant structural abnormalities — including changes in the adjacent temporal bone.
- In light of the TMJ findings, CPAP therapy with mask titration in a sleep lab may be the preferred initial treatment approach.
- If oral appliance therapy is later considered, follow-up imaging and joint function assessment should be completed to ensure safety and minimize risk of joint-related adverse effects.
- Repositioning therapy may be beneficial — particularly avoiding supine sleep and right-side positions, as these were associated with higher respiratory indices.
- Consider lifestyle modifications, including weight reduction and improved sleep hygiene, as supportive strategies for symptom improvement.
- Allergy management and nasal airflow optimization may enhance treatment effectiveness and overall breathing quality.
- A repeat sleep study may be recommended to evaluate treatment response once therapy is established.
- If symptoms persist despite therapy, consider a follow-up DISE to assess upper airway dynamics and guide further treatment adjustments.

Oral Appliance Options for Consideration

Device		Key Features
Emerald Herbst		Strong, lightweight, durable, high-density acrylic for hygiene
Respire Herbst Pink AT		Promotes mandible stabilization, high-density acrylic, metal mesh embedded
Daynaflex Herbst		Accurate fit, enhanced tongue space, stain-resistant PMMA

Disclaimer
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